

Solution Requirements

Date	28 June 2026
Team ID	SWTID-2026-4993
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*)

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Farmers, agricultural experts, and administrators must log in securely using email/mobile number and password to access the system.	High
2	Authorization levels	Admin: Manage users, crops, weather, and system settings. Farmer: View crop recommendations, upload farm details, and receive suggestions. Agricultural Expert: Analyze data and provide crop recommendations and guidance.	High

3	External interfaces	Integrates with weather APIs, soil data sources, market price APIs, and a database to provide real-time crop recommendations and agricultural insights	High
4	Transactions processing	The system collects farmer inputs (crop, soil, location), processes weather and soil data, analyzes it using the optimization engine, and generates crop recommendations. All data is securely stored and updated.	High
5	Reporting	The system generates dashboards and reports showing crop recommendations, soil health, weather forecasts, yield predictions, and farming history. Reports can be downloaded in PDF or Excel format.	High
6	Business rules, etc.	The system generates dashboards and reports showing crop recommendations, soil health, weather forecasts, yield predictions, and farming history. Reports can be downloaded in PDF or Excel format.	High
7	Compliance to laws or regulations	The system follows data privacy and security standards, protects user information, and complies with applicable agricultural and data protection regulations.	Medium

8	Other	The system sends notifications for weather alerts, irrigation schedules, fertilizer recommendations, and crop disease warnings to help farmers make timely decisions.	Medium
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The Flask web application shall process user inputs and return crop predictions within an acceptable response time.	Prediction response in < 2 seconds
2	Scalability	The system shall be designed to support larger agricultural datasets and additional ML models in future versions without major architectural changes.	Supports dataset scaling up to 100,000+ record
3	Security & Data Privacy	All user-submitted agricultural data shall be processed locally and not stored permanently. No personal data is collected or transmitted externally.	No external data transmission; local processing only
4	Reliability & Availability	The application shall run reliably on the Flask local development server without crashes during normal operation and testing.	99% uptime during local testing sessions
5	Usability & Accessibility	The web interface shall be simple, intuitive, and usable by farmers and non-technical users without prior training or technical knowledge.	Usable without training; Bootstrap-based responsive UI
6	Maintainability & Portability	The codebase shall follow modular Python coding standards with clear comments, making it easy to update ML models, add features, or deploy on cloud platforms in future.	Modular code; deployable on AWS/Heroku with minor change

7	Backup & Recovery	The system should regularly back up user data and recover quickly in case of system failure or data loss.	Automatic daily backups and recovery within 30 minutes with no significant data loss.
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